SPCNet: Serial Pyramid Convolutional Network for Remote Sensing Object Detection

Yunping Zheng, Zejun Wang, Kunzhi Wang, Zhou Jiang, and Mudar Sarem

***Abstract*—Recently, remote sensing object detection (RSOD) has attracted increasing attention. Despite advancements in existing methods, challenges like high computational costs, large object scale variations, and difficulties in detecting small objects still remain. Many studies have attempted to address these issues by expanding the reception field, but still struggle with multiscale targets. To address these challenges, we propose serial pyramid convolutional network (SPCNet), an efficient multiscale convolutional network that enhances feature maps by combining serial convolutions with a pyramid structure. First, we have designed a lightweight serial pyramid convolutional (SPC) block that uses depthwise separable convolutions with varying dilation rates to capture multiscale features and a pyramid structure to mitigate feature information loss caused by stacked convolutions. This mechanism is particularly effective at capturing intricate spatial relationships and at focusing on salient regions. In addition, we have developed a novel backbone module, named SPC stage, which integrates multiple SPC blocks along with a parallel large-kernel feed-forward network (FFN) to aggregate multiscale and wide-range contextual information. To confirm its effectiveness, we have evaluated our method on three widely used datasets named HRSC2016, DOTA-v1.0, and DOTA-v1.5. The extensive experiments on these datasets show that our SPCNet achieves the state-of-the-art performance with minimal parameters (72.48% mean average precision (mAP), 78.42% mAP, and 70.60% mAP, respectively). Our code will be made publicly available at: https://github.com/FitzroyWangzj/SPCNet**

***Index Terms*—Lightweight backbone network, multiscale feature, remote sensing object detection (RSOD), sequential convolution.**

I. Introduction

REMOTE sensing object detection (RSOD) is a challenging task aimed at accurately identifying and locating structured objects of interest (such as ships, vehicles, and airplanes) in remote sensing images (RSIs) [1], [2]. This technology has been widely applied in various areas such as traffic monitoring and urban planning. In recent years, significant progress has been made in RSOD tasks in terms of both accuracy and efficiency, thanks to the powerful feature extraction capabilities of deep learning networks.

A substantial body of research has made significant contributions to the design of modules such as encoders, detectors, and loss functions. In terms of detectors, Li et al. [3] designed a multiscale task-enhanced network (MSTEN), which fuses multiscale receptive field features through skip connections and deformable convolutions (DCNs). Tian et al. [4] developed a cross-scale Transformer decoder to achieve information fusion across layers. In addition, Zhang et al. [5] proposed an enhanced strengthen connection feature pyramid network (SC-FPN) that improves lateral connections to better integrate deep and shallow features. For loss functions, Li et al. [6] introduced the elliptical penetrating intersection over union (EPIoU) loss, which leverages inscribed elliptical IoU to enhance localization accuracy. Furthermore, Li et al. [3] proposed a dynamic reppoints learning (DRL) method to address the common issue in current detectors where classification scores do not align with localization accuracy.

Due to the high computational cost, backbone networks specifically designed for RSOD remain underexplored. However, the complex backgrounds and large variations in scale and orientation make generic backbones less effective. Thus, developing a dedicated and lightweight feature extraction backbone is both necessary and practically valuable for remote sensing tasks, especially under limited storage and computational resources.

Recent research on backbones for RSOD has made notable progress. Pu et al. [7] proposed adaptive rotation convolution (ARC), which improves performance by adapting to target orientations but increases computational load and parameters. LSKNet [8] uses large selective kernels for multilevel feature extraction but suffers from high parameter counts and complexity. Similarly, PKINet [9], inspired by InceptionNeXt [10], uses poly kernel inception for multiscale extraction, facing similar challenges in computational cost and parameters.

To address the challenges of multiscale feature extraction and computational constraints, we have proposed a simple yet effective backbone network for RSOD, called the serial pyramid convolutional network (SPCNet). Our network focuses on achieving a large receptive field by sequentially stacking small-kernel depthwise convolution operations. To

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handle the considerable variations in object scale, we have aggregated coarse-to-fine features from multiscale dilated convolutions, enabling multiscale feature fusion while maintaining a lightweight design that delivers strong feature extraction capabilities.

To the best of the authors' knowledge, the SPCNet is the first lightweight stacked convolution-based approach tailored for RSOD. We have conducted experiments on widely used remote sensing benchmark datasets to validate the effectiveness of our method. Our main contributions are as follows.

1) Most current methods use large-kernel convolutions to expand the receptive field, increasing parameter counts. In contrast, SPCNet reduces parameters by stacking smaller convolutions and addresses information loss and feature degradation through feature fusion, offering new insights for future research.
2) We introduce the lightweight SPCBlock, which leverages depthwise dilated convolution and multiscale feature fusion to effectively reduce information loss from stacked convolutions.
3) To validate our approach, we have conducted extensive experiments on the DOTA v1.0 dataset, DOTA v1.5 dataset, and HRSC2016 dataset. The experimental results demonstrate that our model outperforms most recently proposed state-of-the-art models on these datasets with less parameters and FLOPs.

The remainder of this article is organized as follows. Section II introduces and analyzes recent related work. Section III provides a detailed explanation of the structure and implementation of the SPCNet. Section IV presents the experimental results of our model across different datasets. Finally, Section V summarizes our findings and conclusions.

II. Related Works

A. Remote Sensing Object Detection

RSOD aims to accurately identify object locations and orientations in RSIs, facing challenges like scale and orientation variations, background noise, tiny objects, and large image sizes. Traditional methods using horizontal bounding boxes (HBBs) struggle with precise localization, especially for objects with large aspect ratios. This has led to the development of oriented bounding box (OBB) techniques, which better handle arbitrarily oriented targets.

To achieve improved feature extraction and object representation, a variety of solutions had been proposed. Some methods explored oriented object detection frameworks, which could be divided into one-stage and two-stage approaches. One-stage methods, such as RTMDet [11], R3Det [12], and S2ANet [13], directly perform classification and bounding box regression on the input images. Two-stage methods typically use a region proposal network (RPN) to generate candidate regions, followed by classification and bounding box regression on these proposal networks. Examples of these two-stage methods include Faster R-CNN [14], Mask R-CNN [15], Oriented R-CNN [16], and RoI Transformer [17]. Other studies focused on enhancing FPN necks; for instance, Zhang et al. [5] introduced a lateral connection structure to FPN, while Tian et al. [4] proposed FPNFormer, which employed a dual-stream Transformer structure to replace FPN. Some methods also proposed improvements in object representation; for example, Li et al. [3] introduced a flexible adaptive points representation, and Xie et al. [16] developed a midpoint offset representation. Regarding loss functions, Yang et al. [18] improved the smooth $L1$ loss by adding an IoU constant factor, while Zhang et al. [5] used α-CIoU loss to enhance localization accuracy.

In terms of encoders, existing methods have improved performance by introducing various network structures, feature fusion strategies, and optimization techniques. For example, Zhang et al. [5] incorporated ConvNeXt to effectively expand the encoder's receptive field, Pu et al. [7] proposed an ARC method that considered the orientation characteristics of rotated objects. Li et al. [8] introduced large selective kernels to enhance background feature perception, and Cai et al. [9] utilized multiscale convolution operations to improve multiscale feature extraction capabilities.

Recent advances in synthetic aperture radar (SAR) image understanding have also contributed insights relevant to RSOD. To mitigate speckle noise and improve feature quality in SAR imagery, Suo et al. [19] proposed an adaptive enhancement method combining speckle suppression with channelwise fusion. Bo et al. [20] introduced a nuclear norm minus Frobenius norm regularization technique for effective speckle reduction. For SAR-based object detection, Hao and Zhang [21] designed a lightweight CNN tailored for ship detection, and Pu et al. [22] proposed Ulit-BiDet, an ultralightweight SAR object detector based on binary neural networks. These studies underline the growing need for compact yet accurate models in resource-constrained RSOD applications.

Despite of these advanced methods, the research on lightweight backbone networks for RSOD remains limited. In comparison, our proposed SPCNet offers a lightweight architecture that facilitates easier deployment on equipments with limited space.

B. Recent Advances in Lightweight Models

In recent years, with the growing demand for efficient models, research on lightweight networks has gained significant attraction. Iandola et al. [23] proposed the Fire Module, achieving accuracy improvements with a limited number of parameters. ShuffleNet [24] significantly reduced computation costs while maintaining accuracy by employing pointwise group convolution and channel shuffling. Models like MobileNet [25] and Xception [26] reduced parameter count by leveraging depthwise separable convolutions. Tan and Le [27] achieved improved accuracy and efficiency through model scaling. GhostNet [28] addressed feature map redundancy by employing efficient operations, such as depthwise convolution, to capture redundant features while minimizing computational cost.

Some researchers have explored lightweight Vision Transformers (ViTs). Li et al. [29] introduced a new fine-grained joint search algorithm to optimize both the Transformer size and speed. Metha and Rastegari [30] proposed

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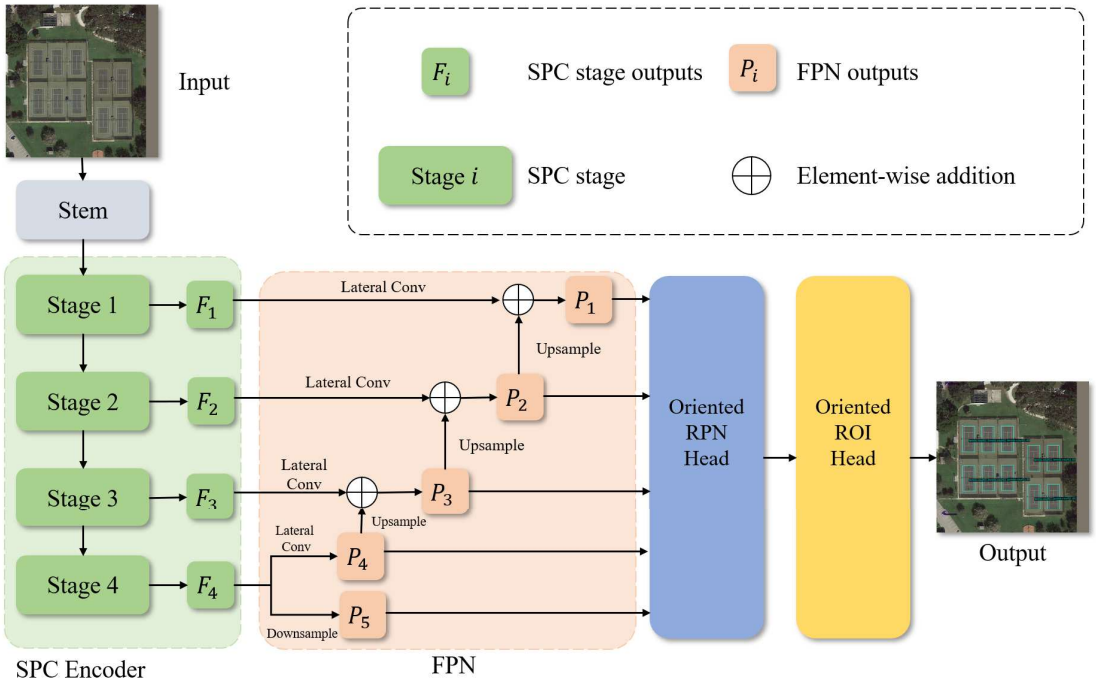


Fig. 1. Overall architecture of SPCNet.

a linear-complexity separable self-attention method, and Pan et al. [31] introduced a cost-effective "local–global–local" (LGL) information exchange bottleneck, achieving a favorable accuracy–efficiency tradeoff. Wang et al. [32] integrated lightweight ViTs into MobileNetV3L. While these models perform well on general vision tasks, further optimization is needed to address the unique characteristics of RSOD.

In response, our SPCNet introduces an innovative approach using small-kernel convolutions and multiscale feature fusion, aiming to enhance RSOD performance while maintaining efficient feature extraction capabilities within a lightweight design.

In summary, although significant progress has been made in RSOD, balancing model complexity, computational efficiency, and feature extraction capability remains a pressing challenge. Our research seeks to address this gap and further advance the field of RSOD.

III. Methods

This article proposes a lightweight RSOD method (SPCNet) based on sequential convolution and pyramid feature aggregation. The overall framework of this method is shown in Fig. 1.

A. Overall Architecture

We have employed our proposed SPC backbone network as the encoder, composed of four SPC stages. Our network features input channel dimensions of $C_{\text{in}} = [32, 64, 128, 256]$ and the number of SPC blocks per stage as $N = [3, 4, 6, 3]$. For an input image $I \in \mathbb{R}^{W \times H \times 3}$, we first process it through a Stem layer for downsampling. This layer consists of a 3×3 convolution kernel with a stride of 2 for downsampling, followed by two 3×3 convolution kernels for feature extraction. The process can be mathematically expressed as follows:

$$F_0 = \text{Conv}_{3\times3}\left(\text{Conv}_{3\times3}\left(\text{Down}\left(I\right)\right)\right) \tag{1}$$

where, Down$(\cdot)$ denotes the downsampling convolution and $\text{Conv}_{3\times3}$ represents a 3×3 convolution. The Stem layer outputs initial feature maps $F_0 \in \mathbb{R}^{C_0 \times H/2 \times W/2}$ with reduced spatial dimensions.

Next, the features are fed into the SPC backbone. For each SPC stage, the feature extraction is performed on input features $F_{i-1} \in \mathbb{R}^{C_i \times H_i \times W_i}$ to produce multiscale features $F_{i+1} \in \mathbb{R}^{C_{i+1} \times H_{i+1} \times W_{i+1}}$, where $C_{i+1} = 2C_i$, $H_{i+1} = H_i/2$, and $W_{i+1} = W_i/2$.

The encoder outputs high-level features that encapsulate rich semantic and contextual information, while the low-level features provide fine-grained details such as local textures and edges. To leverage both features, we integrate a feature pyramid network (FPN) to construct a feature pyramid, enabling the detection of objects across various scales. As illustrated in Fig. 1, the FPN neck comprises lateral connections and a top-down pathway, which progressively aggregates features extracted by the backbone. This process can be expressed as follows:

$$P_i = \text{Lateral}\left(F_i\right) + \text{Up}\left(P_{i+1}\right), \quad i = 1, 2, 3 \tag{2}$$

where P_i represents the ith level feature output by the FPN, Lateral$(\cdot)$ denotes a lateral connection implemented using a 1×1 convolution for channel transformation, and Up$(\cdot)$ represents the upsampling, which adjusts P_{i+1} to match the dimensions of F_i. In addition, the top-level feature undergoes a lateral connection and downsampling to generate two additional features, described as follows:

$$P_i = \text{Lateral}\left(F_i\right), \quad i = 4 \tag{3}$$

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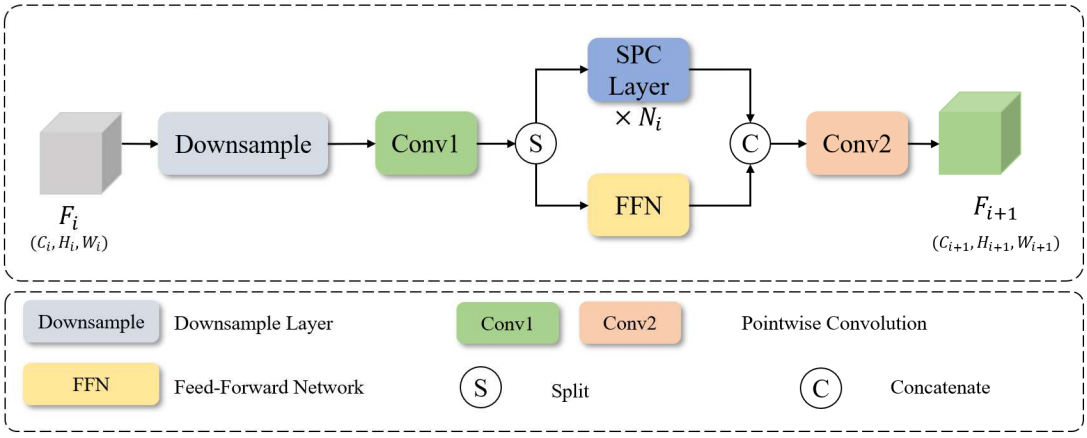


Fig. 2. Architecture of the SPC stage.

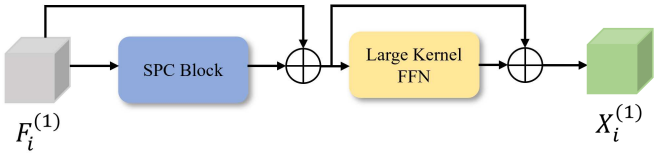


Fig. 3. Technical details of SPC layer.

$$P_{i+1} = \text{Down}\left(F_i\right), \quad i = 4 \tag{4}$$

where Down(·) represents the downsampling, reducing the height and width of the feature maps to half of their original size.

After feature fusion, we utilize the Oriented R-CNN detector for object detection. The Oriented R-CNN is an efficient and accurate two-stage framework for rotated object detection. It comprises an oriented RPN and ROI heads. The RPN head generates rotated bounding box proposals (OBBs), while the ROI head extracts features from these proposals, predicts object categories, and refines bounding box parameters.

B. Serial Pyramid Convolutional (SPC) Stage

The proposed SPCNet backbone is composed of four SPC stages as illustrated in Fig. 2. Each SPC stage incorporates split attention mechanisms, enabling efficient multiscale and diverse feature extraction while maintaining low computational overhead. Initially, the features are downsampled for scale adjustment and channel refinement. Pointwise convolution (PWConv) is then employed to double the channel dimension, allowing for subsequent channel splitting. This splitting operation produces two equal-sized feature maps, $F_i^{(1)}$ and $F_i^{(2)}$, which are processed independently. This operation can be expressed as follows:

$$F_i^{(1)}, F_i^{(2)} = \text{Split}\left(\text{PWConv}\left(F_{i-1}\right)\right), \quad i = 1, 2, 3, 4 \tag{5}$$

where PWConv denotes the pointwise convolution and Split refers to the channel splitting operation.

After splitting, both feature maps are processed heterogeneously. For $F_i^{(1)}$, multiple SPC layers are applied to extract features at various granularity levels. Each SPC layer follows a

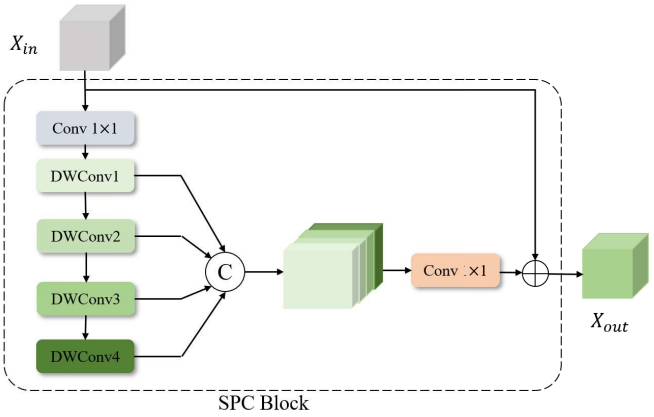


Fig. 4. Architecture of the SPC block.

design paradigm inspired by MetaFormer, leveraging a multi-scale token mixer to extract hierarchical information. Residual connections and feed-forward networks (FFNs) are used to enhance feature representation. To improve the network's adaptability in feature selection and inhibiting noise, learnable adaptive parameters $\gamma_1 \in \mathbb{R}^{1\times1\times C_{i+1}}$ and $\gamma_2 \in \mathbb{R}^{1\times1\times C_{i+1}}$ are introduced within the residual connections. In addition, a novel token mixer, referred to as the SPC block, is proposed to capture both local and global feature interactions. The detailed design and operation of the SPC block are described in Section III-C. The operation of each SPC layer, which is shown in Fig. 3, can be formulated as follows:

$$F_i^{\text{spc}} = F_i^{(1)} + \gamma_1 \cdot \text{SPC}\left(F_i^{(1)}\right), \quad i = 1, 2, 3, 4 \tag{6}$$

$$X_i^{(1)} = F_i^{\text{spc}} + \gamma_2 \cdot \text{FFN}\left(F_i^{\text{spc}}\right), \quad i = 1, 2, 3, 4 \tag{7}$$

where F_i^{spc} denotes the features generated by the SPC block. SPC(·) represents the set of operations performed by the SPC block, which will be discussed in Section III-C. $X_i^{(1)}$ represents the output features of this branch, while FFN(·) denotes a large-kernel FFN. Unlike traditional FFN, we adopt a large-kernel FFN with progressively expanding kernel sizes of 5, 7, 9, and 11 in each stage. This large-kernel convolution design provides a larger receptive field, enabling the network

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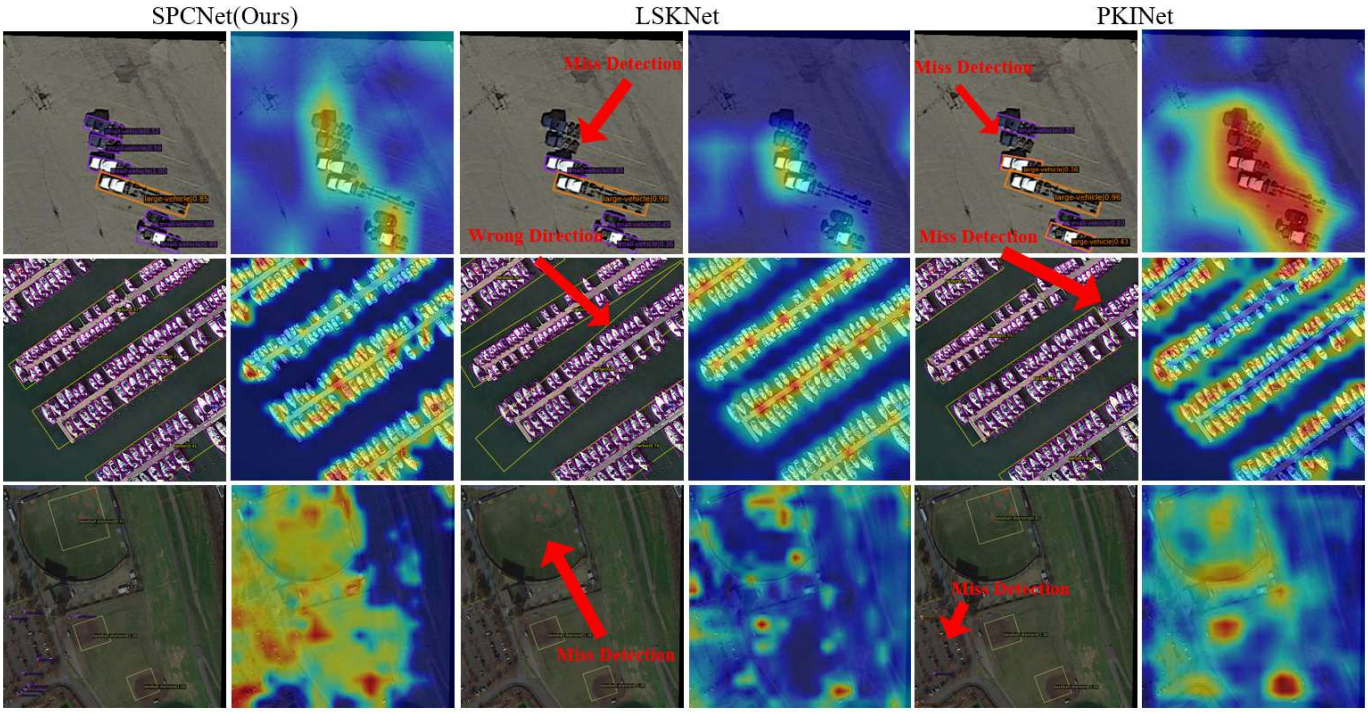


Fig. 5. (Left) Detection result on Dota v1.0 dataset. (Right) Eigen-CAM visualization.

to capture more extensive contextual information and learn more complex feature relationships across different scales. In addition, through residual connections, this module effectively mitigates potential issues such as gradient vanishing and feature degradation.

For $F_i^{(2)}$, a FFN is employed for feature extraction. This lightweight operation complements the information contained in the original feature map.

C. SPC Block

We propose a lightweight SPC Block, the core module of our network (see Fig. 4). Inspired by Inception and Xception, SPC replaces parallel branches with serial ones to enhance interlayer information flow while significantly reducing parameters and computational cost.

However, it is notable that the stacked small-kernel convolutional networks suffer from several limitations, including slow growth of the receptive field, gradient vanishing, and feature degradation, as well as a lack of multiscale feature integration. To address these issues, we have proposed several solutions. First, to tackle the receptive field limitation, we have incorporated dilated convolutions with varying dilation rates instead of large-kernel convolutions, which is computationally expensive. To mitigate the gridding artifacts associated with dilated convolutions, we have used dilation rates of 1, 2, 2, and 3. By employing serial dilated convolutions, we have effectively achieved multiscale receptive fields of 3×3, 7×7, 11×11, and 17×17 at a minimal computational cost, which meets the receptive field requirements for RSOD.

In addition, we have leveraged the depthwise separable convolutions to reduce the number of parameters, further enhancing the model's lightweight design. To address the gradient vanishing and feature degradation, we have introduced a pyramid structure to aggregate the features from different scales by concatenating them, followed by PWConvs to extract multiscale feature information. This facilitates information exchange between the shallow and deep features. The operation can be expressed as follows:

$$X_i = \mathrm{DWConv}_i\left(X_{i-1}\right), \quad i = 1, 2, 3, 4 \tag{8}$$

$$X_{\mathrm{out}} = \mathrm{PWConv}\left(\mathrm{Concat}\left(X_1, X_2, X_3, X_4\right)\right), \quad i = 1, 2, 3, 4 \tag{9}$$

where $\mathrm{DWConv}_i(\cdot)$ represents the ith depthwise separable convolution operation, $\mathrm{Concat}(\cdot)$ denotes the channel concatenation operation, and $\mathrm{PWConv}(\cdot)$ represents the pointwise convolution operation.

IV. Experiments

Our experiments are implemented based on the PyTorch platform and conducted on public datasets such as DOTA v1.0, DOTA v1.5, HRSC2016, SAR Ship Detection Dataset (SSDD), and Vaihingen to evaluate the model performance. The datasets are described in Section IV-A, the implementation details are explained in Section IV-B, the comparative experimental results are presented in Section IV-C, and the ablation study results are demonstrated in Section IV-D.

A. Datasets

DOTA v1.0: DOTA v1.0 is a challenging large-scale RSOD dataset, which contains 2806 RSIs ranging from 800×800 pixels to 4000×4000 pixels. The prediction targets are divided into 15 categories, including plane (PL), baseball

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TABLE I
QUANTITATIVE COMPARISON RESULTS ON THE DOTA V1.0 DATASET USING MAP AS THE CRITERION. THE BEST RESULTS ARE HIGHLIGHTED IN BOLD

Method	Publication	PL	BD	BR	GTF	SV	LV	SH	TC	BC	ST	SBF	RA	HA	SP	HC	mAP ↑
Faster R-CNN [14]	TPAMI2015	89.40	81.81	47.28	67.44	73.96	73.12	85.03	90.90	85.15	84.90	56.60	64.77	64.70	70.28	62.22	73.17
RetinaNet [33]	TPAMI2018	89.29	78.54	41.13	66.29	76.92	61.68	77.40	90.89	81.37	82.89	59.05	63.79	55.05	61.95	40.14	68.42
RoiTrans [17]	CVPR2019	89.01	77.48	51.64	72.07	74.43	77.55	87.76	90.81	79.71	85.27	58.36	64.11	76.50	71.99	54.06	74.05
O-RepPoints [34]	ICCV2019	87.02	83.17	54.13	71.16	80.18	78.40	87.28	90.90	85.97	86.25	59.90	70.49	73.53	72.27	58.97	75.97
FCOS [35]	ICCV2019	88.52	77.54	47.06	63.78	**80.42**	80.50	87.34	90.39	77.83	84.13	55.45	65.84	66.02	72.77	49.17	72.45
SCRDet [36]	ICCV2019	89.98	80.65	52.09	68.36	68.36	60.32	72.41	90.85	**87.94**	86.86	65.02	66.68	66.25	68.24	65.21	72.61
Gliding Vertex [37]	TPAMI2019	89.64	85.00	52.26	77.34	73.01	73.14	86.82	90.74	79.02	86.81	59.55	70.91	72.94	70.86	57.32	75.02
CenterMap [38]	TGRS2020	88.88	81.24	53.15	60.65	78.62	66.55	78.10	88.83	77.80	83.61	49.36	66.19	72.10	72.36	58.70	71.74
S^2ANet [39]	TGRS2020	89.30	80.11	50.97	73.91	78.59	77.34	86.38	90.91	85.14	84.84	60.45	66.94	66.78	68.55	51.65	74.13
SCRDet++ [18]	TPAMI2020	90.05	84.39	55.44	73.99	77.54	71.11	86.05	90.67	87.32	**87.08**	**69.62**	68.90	73.74	71.29	65.08	76.81
R3Det [12]	AAAI2021	89.00	75.60	46.64	67.09	76.18	73.40	79.02	90.88	78.62	84.88	59.00	61.16	63.65	62.39	37.94	69.70
Oriented R-CNN [16]	ICCV2021	89.48	82.59	54.42	72.58	79.01	82.43	88.26	90.90	86.90	84.34	60.79	67.08	74.28	69.77	54.27	75.81
O^2-DETR [40]	Arxiv2021	86.01	75.92	46.02	66.65	79.70	79.93	**89.17**	90.44	81.19	76.00	56.91	62.45	64.22	65.80	58.96	72.15
ReDet [41]	CVPR2021	88.79	82.64	53.97	74.00	78.13	84.06	88.04	90.89	87.78	85.75	61.76	60.39	75.96	68.07	63.59	76.25
CFC-Net [42]	TGRS2021	89.08	80.41	52.41	70.02	76.28	78.11	87.21	90.89	84.47	85.64	60.51	61.52	67.82	68.02	50.09	73.50
AO^2-DETR [43]	TCSVT2022	87.99	79.46	45.74	66.64	78.90	73.90	73.30	90.40	80.55	85.89	55.19	63.62	51.83	70.15	60.04	70.91
SASM [44]	AAAI2022	86.42	78.97	52.47	69.84	77.30	75.99	86.72	90.89	82.63	85.66	60.13	68.25	73.98	72.22	62.37	74.92
DODet [45]	TGRS2022	89.61	83.10	51.43	72.02	79.16	81.99	87.71	90.89	86.53	84.56	62.21	65.38	71.98	70.79	61.93	75.89
CFA [46]	CVPR2023	88.85	75.31	50.68	68.27	79.83	74.61	86.43	90.85	80.67	85.11	50.29	61.05	64.99	67.00	16.65	69.37
ARS-DETR [47]	ICCV2023	86.61	77.26	48.84	66.76	78.38	78.96	87.40	90.61	82.76	82.19	54.02	62.61	72.64	72.80	64.96	73.79
ARC [7]	ICCV2023	89.40	82.48	55.33	73.88	79.37	84.05	88.06	90.90	86.44	84.83	63.63	70.32	74.29	71.91	65.43	77.35
LSKNet [8]	ICCV2023	89.44	84.01	55.45	75.57	79.18	83.70	88.25	90.86	86.08	84.75	63.98	63.68	77.41	75.96	**67.43**	77.71
FANet [48]	TGRS2024	**91.24**	84.42	56.53	68.12	79.82	83.48	88.91	90.56	85.72	86.35	61.96	64.85	74.08	69.14	63.35	76.57
APE [13]	CVPR2024	89.96	53.42	**76.03**	74.01	77.16	79.45	83.62	90.83	87.15	84.51	67.72	**74.61**	71.84	60.33	65.55	75.75
PKINet [9]	CVPR2024	89.38	78.48	55.62	75.04	78.22	82.30	88.05	90.87	87.31	86.10	65.56	69.19	77.41	73.62	62.74	77.74
ORSR [2]	TGRS2024	89.53	85.04	57.20	75.05	77.59	83.90	87.80	90.78	88.87	86.74	63.65	66.30	**78.05**	70.81	69.07	78.03
GSDDet [49]	TGRS2024	88.83	83.34	53.39	75.24	77.83	83.03	88.04	90.86	87.47	85.73	65.25	64.17	75.56	67.62	62.24	76.57
CPMFNet [50]	TGRS2024	89.53	82.92	54.63	77.58	79.37	**84.87**	88.23	**90.95**	86.95	84.93	68.20	70.39	76.35	72.01	66.68	78.23
Ours (SPCNet)	-	88.78	**85.43**	55.45	**79.49**	79.45	83.90	88.18	90.89	85.96	86.97	64.44	69.79	77.30	**79.25**	65.20	**78.70**

diamond (BD), bridge (BR), ground track field (GTF), small vehicle (SV), large vehicle (LV), ship (SH), tennis court (TC), basketball court (BC), storage tank (ST), soccer-ball field (SBF), roundabout (RA), harbor (HA), swimming pool (SP), and helicopter (HC). The dataset is split into training, validation, and testing sets in the proportions of 1/2, 1/6, and 1/3, respectively. The training and validation sets are used for model training, while the testing set is used for analyzing the model's prediction performance. The mean average precision (mAP) and other evaluation metrics are obtained by submitting the prediction results to the official evaluation website. All the images are cropped into 1024 × 1024 patches. To enhance the model's generalization capability, we have applied a random rotation operation with a probability of 50% to the images and normalization.

DOTA v1.5: DOTA v1.5 is an updated version of the DOTA v1.0 dataset. It uses the same images as DOTA v1.0 but includes additional annotations for extremely small instances (less than 10 pixels) and introduces a new category named container crane (CC). The dataset contains a total of 403 318 annotated object instances across 16 categories. The data preprocessing operations are the same as in DOTA v1.0.

HRSC2016: HRSC2016 is a benchmark dataset specifically designed for ship target detection, containing 1061 samples ranging from 300 × 300 pixels to 1500 × 900 pixels. The training set consists of 436 images, while the testing set contains 181 images. Each original image is resized to 800 × 800 pixels.

SSDD: The SSDD is a dataset specifically designed for ship detection in SAR images. It consists of 1160 cropped SAR images with a resolution of approximately 500 × 500 pixels, collected from RadarSat-2, TerraSAR-X, and Sentinel-1 with various polarization modes (HH, HV, VV, and VH). The dataset contains 2456 annotated ship instances in both in-shore and off-shore scenes. All images are resized to 512 × 512 pixels before training, and the training and validation sets are split with a ratio of 8:2.

Vaihingen: Vaihingen is a high-resolution remote sensing semantic segmentation dataset containing 33 true orthophoto (TOP) images with an average size of 2494 × 2064 pixels

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TABLE II
QUANTITATIVE COMPARISON RESULTS ON THE DOTA V1.5 DATASET USING MAP AS THE CRITERION. THE BEST RESULTS ARE HIGHLIGHTED IN BOLD

Method	Publication	PL	BD	BR	GTF	SV	LV	SH	TC	BC	ST	SBF	RA	HA	SP	HC	CC	mAP ↑
RetinaNet [33]	TPAMI2018	71.43	77.64	42.12	64.65	44.53	56.79	73.31	90.84	76.02	59.96	46.95	69.24	59.65	64.52	48.06	0.83	59.16
HTC [51]	CVPR2019	77.80	73.67	51.40	63.99	51.54	73.31	80.31	90.48	75.12	67.34	48.51	70.63	64.84	64.48	55.87	5.15	63.40
Faster R-CNNN [14]	TPAMI2015	71.89	74.47	44.45	59.87	51.28	68.98	79.37	90.78	77.38	67.50	47.75	69.72	61.22	65.28	60.47	1.54	62.00
Mask R-CNN [15]	ICCV2017	76.84	73.51	49.90	57.80	51.31	71.34	79.75	90.46	74.21	66.07	46.21	70.61	63.07	64.46	57.81	9.42	62.67
ReDet [41]	CVPR2021	79.20	82.81	51.92	71.41	52.38	75.73	80.92	90.83	75.81	68.64	49.29	72.03	73.36	70.55	**63.33**	11.53	66.86
LSKNet [8]	ICCV2023	72.05	**84.94**	**55.41**	**74.93**	**52.42**	**77.45**	81.17	**90.85**	**79.44**	69.00	**62.10**	**73.72**	**77.49**	**75.29**	55.81	**42.19**	70.26
Ours (SPCNet)	-	**79.82**	84.25	53.78	74.24	52.15	76.66	**87.35**	**90.85**	78.65	**69.45**	61.95	72.98	76.11	72.96	61.25	37.18	**70.60**

and a ground sample distance (GSD) of 9 cm. The dataset covers six semantic categories: impervious surface, building, low vegetation, tree, car, and background (clutter). Following standard practice, a subset of images is used as the test set (e.g., IDs 2, 4, 6, …, 38), while the rest form the training set. For model training, images are cropped into patches of 512 × 512 or 256 × 256 pixels depending on the setup.

B. Implementation Details

All experiments are performed on the NVIDIA GeForce RTX 3090 GPU. In addition, we have built the experimental environment based on PyTorch and the MMRotate framework. For the pretraining process, the AdamW optimizer is adopted with a learning rate of 0.0001, a momentum set to 0.9, a weight decay set to 0.05, and a batch size of 64 for 100 epochs to reduce the training time. In the subsequent training phase, we have employed the AdamW optimizer with the learning rate initialized to 0.0001, betas set to 0.9 and 0.999, and the weight decay of 0.05. For remote sensing detection datasets, we adopted Oriented R-CNN as the detector and trained our model for 30 epochs to ensure sufficient convergence. For remote sensing segmentation datasets, we adopted UNetFormer [52] as the decoder and trained our model for 104 epochs.

C. Comparisons With State of the Arts

To verify the effectiveness of our model, we have compared our model with the state-of-the-art models on several remote sensing detection and segmentation datasets.

1) Performance on DOTA v1.0: Table I demonstrates a comparison of the prediction performance between our model and the state-of-the-art models, from which we can find that our model exceeds the previously best backbone PKINet by 0.96% in mAP. Furthermore, our model achieves breakthrough performance improvements on some challenging objectives, such as ground track fields and swimming pools, with enhancements of 4.61% and 2.36%, respectively, compared with the best previous results. In addition, compared with the Oriented R-CNN, which uses the same detector, our model shows a 2.61% improvement in prediction performance. These experimental results demonstrate the effectiveness of the SPCNet we have proposed. Fig. 5 shows the detection results of our model. It can be observed that our model can effectively tackle various challenges, such as low contrast, multiple object combinations, and distinguishing similar objects. In addition,

TABLE III
QUANTITATIVE COMPARISON RESULTS ON THE HRSC2016 DATASET USING MAP AS THE CRITERION. THE BEST RESULTS ARE HIGHLIGHTED IN BOLD

Method	Publication	mAP ↑
RetinaNet [33]	TPAMI2018	52.70
S^2ANet [39]	TGRS2020	55.65
Oriented-RCNN [16]	ICCV2021	70.55
PKINet [9]	CVPR2024	72.11
ARC [7]	ICCV2023	72.39
Ours (SPCNet)	-	**72.48**

we employ Eigen-CAM [53] to visualize the model results (as shown in Fig. 5). It can be observed that when there are large-scale differences among target objects, most models tend to focus on the main target while neglecting fine-grained details. In contrast, our model leverages stacked multiscale receptive field information to better capture these details, thereby enhancing its localization capability.

2) Performance on DOTA v1.5: The results of comparison experiments are shown in Table II. Our model outperforms most models with an mAP of 70.60%. Moreover, our model has more stable prediction performance across various categories compared with other models. Due to the significant disadvantage in the number of parameters, the overall performance of our model is excellent. We also notice that the model underperforms on certain categories. We believe that the relatively lower performance in certain categories is mainly due to the significant class imbalance, as some categories, such as container crane, contain only 283 samples, respectively, accounting for less than 0.3% of the total.

3) Performance on HRSC2016: Table III represents the evaluation results of our model and other models on the HRSC2016 dataset. Our model achieves the best performance with a 72.48% mAP despite having a smaller number of parameters.

4) Performance on SSDD: We evaluated the performance of our proposed SPCNet on the SSDD dataset and compared it against six state-of-the-art object detection methods, as shown in Table IV. The results clearly demonstrate the superior performance of our SPCNet, which achieves an mAP of 81.12%, surpassing all competing methods. Notably, it outperforms the recent PKINet and Oriented R-CNN by margins of 0.19% and 0.20%, respectively.

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TABLE IV
QUANTITATIVE COMPARISON RESULTS ON THE SSDD DATASET USING MAP AS THE CRITERION. THE BEST RESULTS ARE HIGHLIGHTED IN BOLD

Method	Publication	mAP ↑
RetinaNet [33]	TPAMI2018	71.20
FCOS [35]	ICCV2019	78.67
ReDet [41]	CVPR2021	79.73
Oriented-RCNN [16]	ICCV2021	80.92
LSKNet [8]	ICCV2023	78.79
PKINet [9]	CVPR2024	80.93
Ours (SPCNet)	-	**81.12**

TABLE V
QUANTITATIVE COMPARISON RESULTS ON THE VAIHINGEN DATASET

Model	mF1 ↑	OA ↑	mIOU ↑
PSPNet [54]	79.0	87.7	68.6
DANet [55]	79.6	88.2	69.4
SwiftNet [56]	88.3	90.2	79.6
UNetFormer [52]	90.4	91.0	82.7
LSKNet [8]	90.6	**93.5**	83.2
PKINet [9]	90.5	93.2	83.0
Ours(SPCNet)	**90.8**	93.2	**83.5**

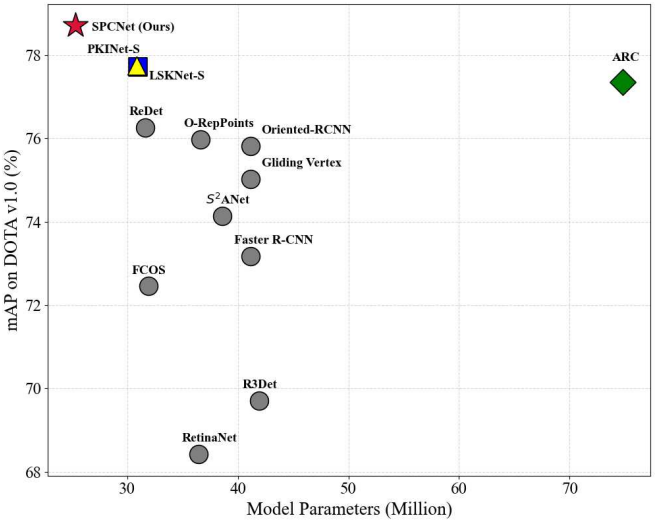


Fig. 6. Model size versus detection performance.

5) Performance on Vaihingen: Table V shows the evaluation results on the Vaihingen dataset. It can be observed that our model still demonstrates impressive performance on the semantic segmentation task.

6) Model Parameters and Reference Speed: The model's parameter scale and inference speed represent spatial efficiency and temporal efficiency, respectively. Consequently, we have compared our SPCNet with several mainstream models in terms of parameter scale and FLOPs. All backbones are equipped with an FPN neck and Oriented R-CNN as the detector. The comparison results are shown in Table VI and Fig. 6. Compared with the previous best-performing model, PKINet, our model reduces the parameter count by 5.57M (25.29M versus 30.86M) and decreases the FLOPs by 31.24G (153.2G versus 184.44G). Regarding training time, although our model does not have an advantage compared with models like LSKNet, it achieves significant improvements over the baseline model (PKINet). This clearly demonstrates that the serial depthwise convolutions and pyramid structures effectively reduce the feature redundancy and enhance the model efficiency.

TABLE VI
COMPARISON OF MODEL TRAINING TIME, PARAMETERS, AND FLOPS. AVE. ITER. TIME: AVERAGE ITERATION TIME

Model	Publication	Ave. Iter. Time	Parameters ↓	FLOPs ↓
Oriented-RCNN [16]	ICCV2021	0.4288s	41.14M	206.81G
LSKNet [8]	ICCV2023	**0.3690s**	30.98M	173.59G
ARC [7]	ICCV2023	0.4673s	74.83M	176.48G
PKINet [9]	CVPR2024	1.6096s	30.86M	184.44G
Ours(SPCNet)	-	0.4737s	**25.29M**	**153.2G**

TABLE VII
ABLATION STUDIES OF THE DEPTHWISE CONVOLUTION MODULE ON DOTA V1.0 DATASET. $d = n$ MEANS THE DILATION RATE IS n

Model	Parameters ↓	FLOPs ↓	$d=1$	$d=2$	$d=3$	mAP
SPCNet	24.95M	151.98G		✓	✓	77.87
SPCNet	24.95M	151.98G	✓		✓	77.37
SPCNet	24.95M	151.98G	✓	✓		76.39
Ours(SPCNet)	25.29M	153.2G	✓	✓	✓	**78.42**

D. Ablation Experiments

In our model, the serial pyramid convolution module is a key for expanding the receptive fields and enhancing the features. Therefore, in the ablation experiments, we validate whether the changes in the receptive field and the reduction in the feature information would lead to a decline in the model performance by removing certain deep convolution modules. As shown in Table VII, we have found that removing some convolution modules significantly reduced the model's performance. Moreover, dropping convolutions with larger dilations results in a greater performance drop. This indicates that the receptive field and the deep semantic features are crucial factors affecting the accuracy of the RSOD.

V. CONCLUSION

In this article, we propose SPCNet, a lightweight backbone designed for RSOD. By combining serial pyramid convolutions with multiscale dilated depthwise separable convolutions, SPCNet achieves a large receptive field with high efficiency. The SPC block and pyramid feature fusion effectively reduce information loss and enhance multiscale feature extraction. Experiments on DOTA v1.0, DOTA v1.5, and HRSC2016 show that SPCNet outperforms state-of-the-art methods with fewer parameters and FLOPs. Ablation studies confirm the effectiveness of the proposed components in balancing performance and efficiency.

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Despite its strengths, SPCNet may struggle with detecting extremely small objects in complex backgrounds—a common challenge in remote sensing. Future work will explore dynamic attention, hybrid Transformer architectures, adaptive dilation, and lightweight cross-scale fusion to enhance robustness and generalization. With advances in lightweight models and large-scale datasets, SPCNet offers a solid foundation for efficient, accurate real-world object detection.

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